

Evaluation Frameworks and RAG Triad Metrics

The RAG Triad Metrics

Evaluating RAG systems requires measuring three orthogonal dimensions: Context Relevance (is retrieved context pertinent?), Groundedness (is response supported by context?), and Answer Relevance (does response answer query?).

Automated LLM-as-a-Judge

Frameworks like Ragas and TruLens employ GPT-4 or Claude as an evaluator to score prompt-context alignment and detect hallucinations systematically.

IR Ground-Truth Benchmarking

Retrieval performance is benchmarked against gold snippets using Recall@K, Precision@K, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG).